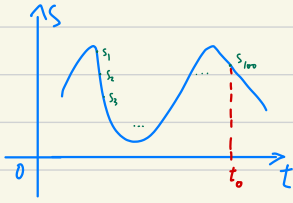


Definition



• $\mathbb{R}^N \mapsto \mathbb{R}$ (left panel)

$$\cdot \quad r = r(s_1, \dots, s_N) \quad (s_1, \dots, s_N) \mapsto r(t_0)$$

$$\cdot \quad r(\vec{s}_0 + \delta \vec{s}) = r(\vec{s}_0) + \sum_{i,j} \frac{\partial r}{\partial s_i} \delta s_i + \sum_{i,j,k} \frac{\partial^2 r}{\partial s_i \partial s_j} \delta s_i \delta s_j + \sum_{i,j,k,l} \frac{\partial^3 r}{\partial s_i \partial s_j \partial s_k \partial s_l} \delta s_i \delta s_j \delta s_k \delta s_l + \dots$$

• $\mathbb{R}^{+\infty} \mapsto \mathbb{R}$ (right panel)

$$\cdot \quad r = r(s(t)) \quad s(t) \in (t_0, t_1) \mapsto r(t_0)$$

$$\cdot \quad r(s_0 + \delta s) = r(s_0) + \int dt \frac{\delta r}{\delta s(t-z)} \delta s(t-z) + \iint dt dz \frac{\delta^2 r}{\delta s(t-z) \delta s(t-z')} \delta s(t-z) \delta s(t-z') + \iiint dt dz dz' \frac{\delta^3 r}{\delta s(t-z) \delta s(t-z') \delta s(t-z'')} \delta s(t-z) \delta s(t-z') \delta s(t-z'') + \dots$$

where $\left\{ \begin{array}{l} \frac{\delta r}{\delta s(t-z)} \stackrel{\Delta}{=} \lim_{\Delta t \rightarrow 0} \frac{(\frac{\partial r}{\partial s_i})}{\Delta t} \\ \frac{\delta^2 r}{\delta s(t-z) \delta s(t-z')} \stackrel{\Delta}{=} \lim_{\Delta t \rightarrow 0} \left(\frac{\partial^2 r}{\partial s_i \partial s_j} \right) \frac{1}{(\Delta t)^2} \\ \frac{\delta^3 r}{\delta s(t-z) \delta s(t-z') \delta s(t-z'')} \stackrel{\Delta}{=} \lim_{\Delta t \rightarrow 0} \left(\frac{\partial^3 r}{\partial s_i \partial s_j \partial s_k} \right) \frac{1}{(\Delta t)^3} \\ \dots \end{array} \right.$ // We assume that the R.H.S. is the same for app t:

• Above, we introduce definition 1: $\frac{\delta r}{\delta s(t-z)} \stackrel{\Delta}{=} \lim_{\Delta t \rightarrow 0} \frac{(\frac{\partial r}{\partial s_i})}{\Delta t}$

We can expand it as below to give the testing function definition:

$$\cdot \quad \frac{\delta r}{\delta s(t-z)} \stackrel{\Delta}{=} \lim_{\Delta t \rightarrow 0} \frac{(\frac{\partial r}{\partial s_i})}{\Delta t} = \lim_{\Delta t \rightarrow 0} \lim_{\varepsilon \rightarrow 0} \frac{r(s_1, \dots, s_i + \varepsilon, \dots) - r(s_1, \dots, s_i, \dots)}{\varepsilon} \stackrel{\Delta}{=} \lim_{\Delta t \rightarrow 0} \lim_{\varepsilon \rightarrow 0} \frac{r(s_1, \dots, s_i + \frac{\varepsilon}{\Delta t}, \dots) - r(s_1, \dots, s_i, \dots)}{\frac{\varepsilon}{\Delta t}}$$

• We use $\varepsilon' \delta(t-z)$ to replace $\frac{\varepsilon}{\Delta t}$, since they are 0.00 and % responsivity, then both are a small perturbation

$$\text{so } \frac{\delta r}{\delta s(t-z)} = \lim_{\varepsilon \rightarrow 0} \frac{r(s_1, \dots, s_i + \varepsilon' \delta(t-z), \dots) - r(s_1, \dots, s_i, \dots)}{\varepsilon'}$$

• We assume that the R.H.S. is the same for app t: , so $\frac{\delta r}{\delta s(t-z)} = \lim_{\varepsilon \rightarrow 0} \frac{r(s(t) + \varepsilon' \delta(t-z)) - r(s(t))}{\varepsilon'}$

• We assume steadyness, so $\frac{\delta r}{\delta s(t-z)} = \frac{\delta r}{\delta s(t-z)} = \lim_{\varepsilon \rightarrow 0} \frac{r(s(t) + \varepsilon' \delta(t-z)) - r(s(t))}{\varepsilon'}$

• In summary $\left\{ \begin{array}{l} \text{def 1: } \frac{\delta r}{\delta s(t-z)} \stackrel{\Delta}{=} \lim_{\Delta t \rightarrow 0} \frac{(\frac{\partial r}{\partial s_i})}{\Delta t} \\ \text{def 2: } \frac{\delta r}{\delta s(t-z)} = \lim_{\varepsilon \rightarrow 0} \frac{r(s(t) + \varepsilon' \delta(t-z)) - r(s(t))}{\varepsilon'} \end{array} \right.$

• We use heuristic methods to get the 2 defs. If you're unhappy, just accept these as defs. Leave the mathematician to rigorize them!

Clarification

map

- $\mathbb{R} \mapsto \mathbb{R} : y = f(x)$
- $\mathbb{R}^n \mapsto \mathbb{R}^m : \vec{y} = f(\vec{x}_{n \times 1})$
- $\mathbb{R}^{n_1 \times n_2} \mapsto \mathbb{R}^{m_1 \times m_2} : Y_{m_1 \times m_2} = f(X_{n_1 \times n_2})$
- $\mathbb{R}^n \mapsto \mathbb{R} : y = f(\vec{x}_n)$
- $\mathbb{R}^{+\infty} \mapsto \mathbb{R} : y = F(f_{1:n}) \quad \text{or} \quad f(x) \mapsto y$

intuitive example

$$C = \pi \cdot D$$

feature $\mapsto$ single label

PCA

feature $\mapsto$ multiple label

Maximize S , subject to C ; The Quickest Curve

notation

- $y = F(f_{1:n})$ or $y = g(f_{1:n})$ is not a good notation, causing confusion. I recommend use $\mathbb{R} \rightarrow \mathbb{R}$ or $x \mapsto f(x)$ for $f(x)$,
 $\mathbb{R} \mapsto \mathbb{R} \mapsto \mathbb{R}$ or $x \mapsto f(x) \mapsto g(f(x))$ for $y = g(f(x))$ and $\mathbb{R}^{+\infty} \mapsto \mathbb{R}$ or $f_{1:n} \mapsto y$ for $y = F(f_{1:n})$

- $\frac{\delta r}{\delta s(t)} \triangleq \lim_{\delta t \rightarrow 0} \left(\frac{\delta r}{\delta s(t)} \right) \frac{1}{\delta t}$ // We assume that the R.H.S. is the same for all t :

I recommend $\mathbb{R}^{+\infty} \mapsto \mathbb{R}$ or $s(t) \in (t_0, t_1) \mapsto r(t_0)$ for $t = t(s(t))$
and $\mathbb{R} \mapsto \mathbb{R}$ or $t_0 \mapsto \frac{\delta r}{\delta s(t)}(t_0)$ for $\frac{\delta r}{\delta s(t)}$

- from finite to infinite: We can see functional as $\mathbb{R}^n \rightarrow \mathbb{R} \quad (n \rightarrow \infty)$
see functional derivative = 0 as $+\infty$ partial derivative = 0

Example

- Given $E \triangleq \int f(x) p_n f(x) dx$, calculate $\frac{\delta E}{\delta f(x)}$

$$E = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) p_n f(x_i) \Delta x \Rightarrow \forall x_i: \frac{\partial E}{\partial f(x_i)} = \Delta x (p_n f(x_i) + 1) \Rightarrow \frac{\delta E}{\delta f(x)} = p_n f(x) + 1$$

$$\frac{\delta E}{\delta f(x)} \triangleq \frac{\partial E}{\partial f(x)} \frac{1}{\Delta x} \quad (\Delta x \rightarrow 0)$$

- Given $E \triangleq \int f(x) p_n f(x) dx$, Prove $\frac{\delta E}{\delta f(x)} = 0 \Leftrightarrow p_n f(x) + 1 = 0$

$$\frac{\delta E}{\delta f(x)} = 0 \Leftrightarrow \forall x_i: \frac{\partial E}{\partial f(x_i)} = 0 \Leftrightarrow \forall x_i: \Delta x (p_n f(x_i) + 1) = 0 \Leftrightarrow \forall x_i: p_n f(x_i) + 1 = 0 \Leftrightarrow p_n f(x) + 1 = 0$$

- When we use Dirac function to test (you can use other functions), i.e. $\frac{\delta r}{\delta s(t)} \triangleq \lim_{\varepsilon \rightarrow 0} \frac{r(s(t) + \varepsilon \delta(t-t_0)) - r(s(t))}{\varepsilon}$ calculate below functional derivative

$$E = \int f(x) dx \Rightarrow \frac{\delta E}{\delta f(x)} = \lim_{\varepsilon \rightarrow 0} \frac{\int (f(x) + \varepsilon \delta(x-x')) dx - \int f(x) dx}{\varepsilon} = \int \delta(x-x') dx = 1$$

$$E = \int f(q(t)) dt \Rightarrow \frac{\delta E}{\delta q} = \frac{dE}{dt} \frac{dt}{dq} = 1 \cdot \frac{dt}{dq} = \frac{dt}{dq}$$

$$S = \int L(q(t), \dot{q}(t), t) dt \Rightarrow \frac{\delta S}{\delta q(t')} = \lim_{\varepsilon \rightarrow 0} \frac{\int L(q + \varepsilon \delta(t-t'), \dot{q} + \varepsilon \delta'(t-t'), t) dt - \int L dt}{\varepsilon} \stackrel{\text{Taylor, 保留一阶}}{=} \lim_{\varepsilon \rightarrow 0} \frac{\int [L + \varepsilon \frac{\partial L}{\partial q} \delta(t-t') + \varepsilon \frac{\partial L}{\partial \dot{q}} \delta'(t-t')] dt - \int L dt}{\varepsilon}$$

$$= \int \left(\frac{\partial L}{\partial q} \delta(t-t') + \frac{\partial L}{\partial \dot{q}} \delta'(t-t') \right) dt$$

$$= \frac{\partial L(q, \dot{q}, t')}{\partial q} + \int_a^b dt \left[\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \delta(t-t') \right) - \delta(t-t') \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) \right]$$

$$= \frac{\partial L(q, \dot{q}, t')}{\partial q} - \frac{d}{dt'} \left(\frac{\partial L(q, \dot{q}, t')}{\partial \dot{q}} \right) + \left[\frac{\partial L}{\partial \dot{q}} \delta(t-t') \right]_a^b$$

↓ $a < t' < b$ then this term = 0 - 0 = 0

$$= \frac{\partial L(q, \dot{q}, t')}{\partial q} - \frac{d}{dt'} \left(\frac{\partial L(q, \dot{q}, t')}{\partial \dot{q}} \right)$$

$$E = \frac{\delta f(x)}{\delta f(y)} \Rightarrow E = \frac{\delta}{\delta f(y)} \int f(z) \delta(x-z) dz = \lim_{\varepsilon \rightarrow 0} \frac{\int (f(z) + \varepsilon \delta(z-y)) \delta(x-z) dz - \int f(z) \delta(x-z) dz}{\varepsilon} = \int \delta(z-y) \delta(x-z) dz = \delta(x-y)$$

More detailed: $\begin{cases} f(t_0) \triangleq R(s(t)) & t \in (-\infty, t_0] \\ f(t_0) = s(t_0) & t \in (t_0, \infty) \end{cases} \xrightarrow{R} f(t_0) \Rightarrow \forall z \in (-\infty, t_0) \quad \frac{\delta f(t_0)}{\delta s(z)} \stackrel{\text{def}}{=} \frac{\delta R(s(t))}{\delta s(z)} \quad t \in (-\infty, t_0]$

$$\stackrel{\text{def}}{=} \frac{R(s(t_0) + \varepsilon \delta(t_0-z)) - R(s(t_0))}{\varepsilon} \quad t \in (-\infty, t_0] = \frac{(s(t_0) + \varepsilon \delta(t_0-z)) - s(t_0)}{\varepsilon} = \frac{\varepsilon \delta(t_0-z)}{\varepsilon} = \delta(t_0-z)$$

- t_0 is when the response is, z is when the testing function is
- IR and function $\begin{cases} \text{If you fix } z \text{ and } t_0, \text{ then } \frac{\delta f(t_0)}{\delta s(z)} \text{ and } \delta(t_0-z) \text{ are both } \in \mathbb{R} \\ \text{If you change } z \text{ and } t_0, \text{ then } \frac{\delta f(t_0)}{\delta s(z)} \text{ and } \delta(t_0-z) \text{ are both } \in \text{Function} \end{cases}$

Example of Hubel and Wiesel

- Given $\begin{cases} r_{\text{estimate}}(t) = r_0 + \frac{1}{T} \int_0^T D(z) s(t-z) dz \\ E_{\text{error}} = \frac{1}{T} \int_0^T (r(t) - r_{\text{estimate}}(t))^2 dt \end{cases}$ where s is white noise and D is called kernel

Calculate the kernel that minimize Error

'P.S.: $\frac{1}{T}$ is optional, adding it can correct the dimension)

- $\frac{\delta E}{\delta D(z)} = 0 \Rightarrow \int_0^T dt (r_{\text{estimate}}(t) - r(t)) \frac{\delta r_{\text{estimate}}}{\delta D(z)} = 0 \Rightarrow \int_0^T dt (r_{\text{estimate}}(t) - r(t)) s(t-z) = 0$

$$\Rightarrow \int_0^{+\infty} dz' D(z') Q_{ss}(t-z') = Q_{rs}(t-z)$$

$$\Rightarrow \int_0^{+\infty} D(z') dz' \delta(t-z') \sigma^2 = Q_{rs}(t-z)$$

$$\Rightarrow D(z) = \frac{1}{\sigma^2} Q_{rs}(1-z)$$

- $\begin{cases} D(z) = \frac{1}{\sigma^2} Q_{rs}(1-z) \\ C(z) = \frac{1}{\sigma^2} Q_{rs}(1-z) \end{cases} \Rightarrow D(z) = \frac{1}{\sigma^2} \cdot \langle r \rangle \cdot C(z)$ app can be measured

- Dimension $\begin{cases} Q_{rs}(z) : [r] \cdot [s] \\ C(z) : [s] \\ D(z) : [r] [s]^{-1} \end{cases}$